

Centre Number	Candidate Number	Candidate Name
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NAMIBIA SENIOR SECONDARY CERTIFICATE

AGRICULTURAL SCIENCE ORDINARY LEVEL

6115/1

PAPER 1

2 hours

Marks 120

2023

Additional Material: Non-programmable calculator

INSTRUCTIONS AND INFORMATION TO CANDIDATES

- Candidates answer on the Question Paper in the spaces provided.
- Write your Centre Number, Candidate Number and Name in the spaces at the top of this page.
- Write in dark blue or black pen.
- You may use a soft pencil for any diagrams, graphs or rough working.
- Do not use correction fluid.
- You may use a non-programmable calculator.
- Do not write in the margin *For Examiner's Use*.
- The number of marks is given in brackets [] at the end of each question or part question.

Section A

- Answer **all** questions.

Section B

- Answer any **two** questions.
- Write your answers on the answer sheets at the back of the booklet.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
8	
Section B	
.....	
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Total	

Marker	
Checker	

This document consists of **21** printed pages and **3** blank pages.



Republic of Namibia
MINISTRY OF EDUCATION, ARTS AND CULTURE

SECTION A

Answer **all** questions in the spaces provided.

1 (a) Fig. 1.1 shows a pot plant.

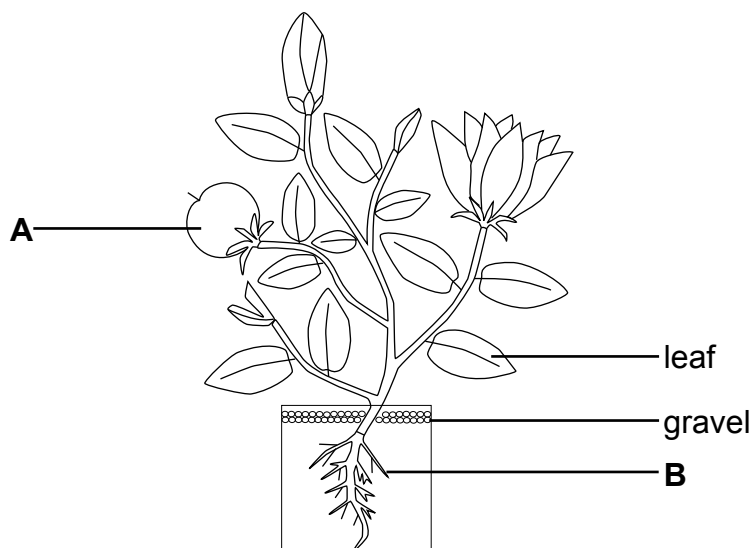


Fig. 1.1

- (i) Name the macro nutrients needed by the plant for the development of the parts labelled **A** and **B**.

A.....

B.....

[2]

- (ii) Describe **two** symptoms of nutrient deficiency that may be shown by a leaf.

1

.....

2

.....

[2]

- (iii) Explain how the plant in Fig. 1.1 can be affected negatively if the temperature drops below 0°C.

.....

.....

.....

.....

[3]

- (iv) Suggest **one** advantage and **one** disadvantage of placing gravel on the soil surface in the plant pot as shown in Fig. 1.1.

advantage

disadvantage [2]

- (b) Explain how over irrigation could contribute to low pH in a soil.

.....

 [3]

- (c) Fig. 1.2 shows a rainfall graph.

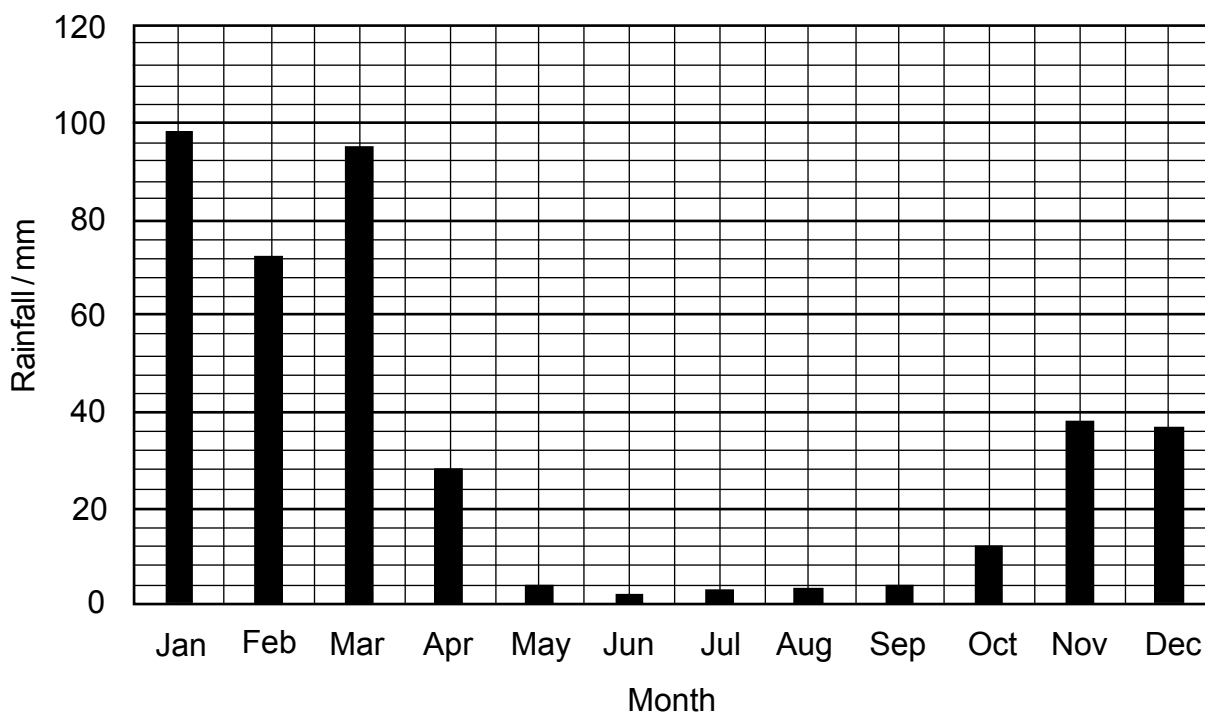


Fig. 1.2

Calculate the average rainfall for the first 3 months of the year.

[2]

[14]

- 2 Fig. 2.1 shows the distribution of soil components in a poorly drained, waterlogged soil.

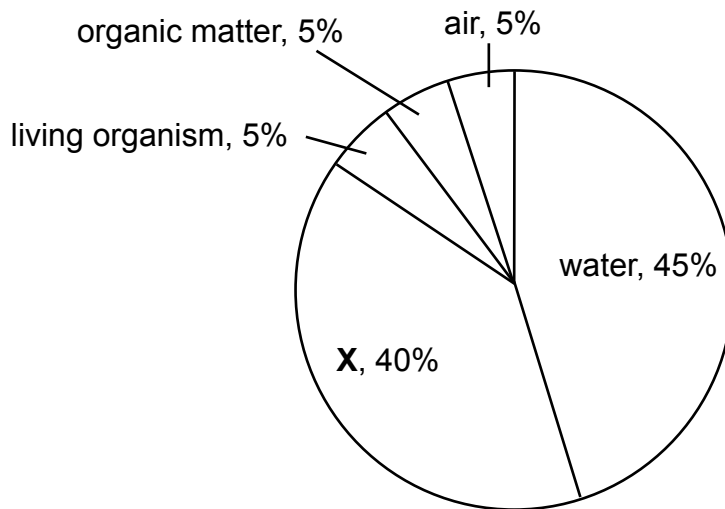


Fig. 2.1

- (a) State **two** methods used to drain a waterlogged soil.

1.....

2.....

[2]

- (b) (i) Name the soil component labelled **X**.

.....

[1]

- (ii) Work out the percentage of pore spaces that is occupied by water.

[2]

- (iii) Describe the role of organic matter in the soil.

.....

[4]

- (iv) Suggest how plant growth might be affected when grown in the soil indicated by Fig. 2.1.

.....

.....

.....

.....

.....

.....

[3]

[12]

- 3 A crop plant was grown in a soil recently flooded by the sea, leaving the soil with a high salt concentration. As a result its leaves started drying up, eventually falling off.

(a) Describe and **explain** the processes that take place in the plant cells leading to the drying of leaves.

.....

.....

.....

.....

.....

[3]

(b) (i) Explain how a reduced number of leaves will affect photosynthesis.

.....

.....

.....

[2]

(ii) Discuss the importance of photosynthesis to plants.

.....

.....

.....

[2]

(c) (i) Define *translocation*.

.....

.....

.....

[2]

(ii) Describe the modification of **one** plant part into a food storage organ and name the type of food material it stores.

.....

.....

.....

[2]

(d) Which crop reproduce asexually?

- A bean
- B maize
- C sorghum
- D Irish potato

Answer **A, B, C** or **D** [1]

[12]

4 Fig. 4.1 shows two common weeds affecting crops in Namibia.

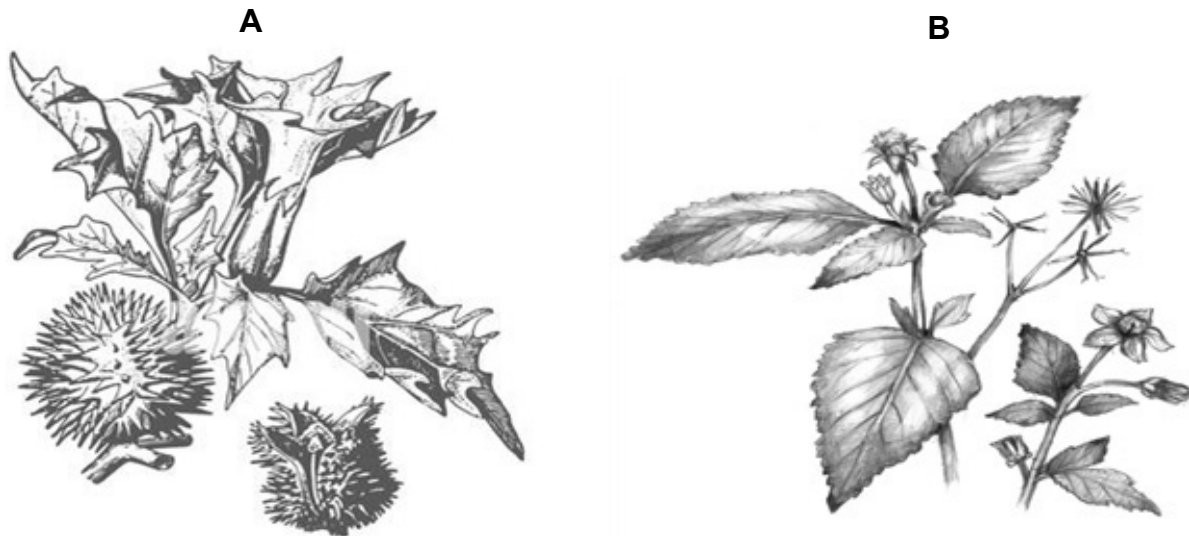


Fig. 4.1

(a) (i) Identify weeds **A** and **B** in Fig. 4.1.

A.....

B..... [2]

(ii) Suggest the mode of spread of weed **B**.

.....

..... [1]

(b) Fig. 4.2 shows some harmful effects of weeds on crops.

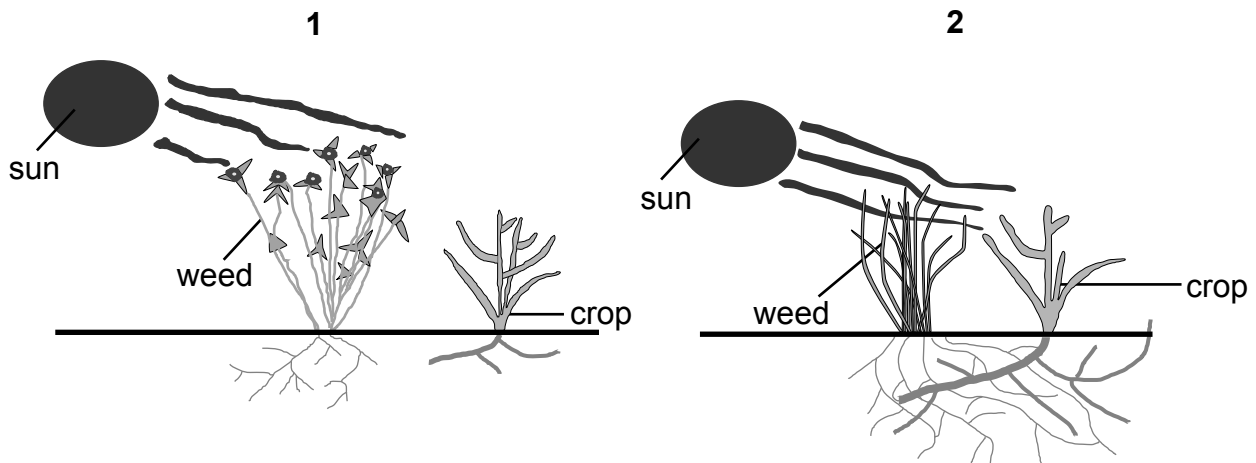


Fig. 4.2

Which harmful effects of weeds are illustrated in Fig. 4.2 at **1** and at **2**?

1.....

2..... [2]

(c) Describe how weeds are controlled using

biological control,.....

.....

.....

chemical control.

.....

.....

[4]

(d) Describe what is meant by Integrated Pest Management.

.....

.....

.....

.....

.....

[3]

(e) Which type of disease is shown in Fig. 4.3?

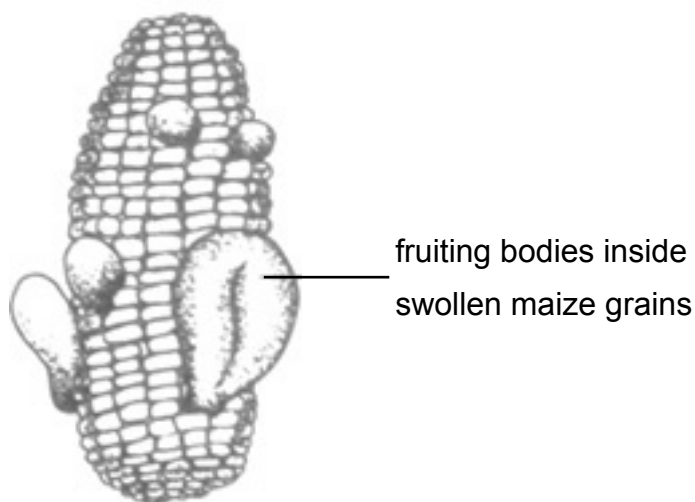


Fig. 4.3

- A** blight
- B** wilt
- C** smut
- D** streak

Answer **A, B, C** or **D** [1]

[13]

5 Fig. 5.1 shows a lactating cow.

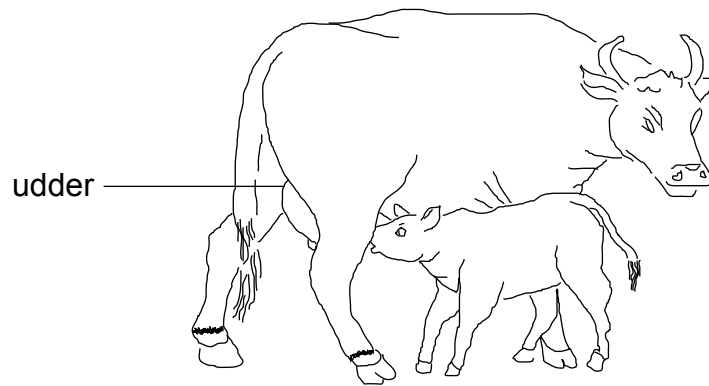


Fig. 5.1

(a) (i) State what is meant by *lactation*.

.....
 [1]

(ii) The cow produces colostrum immediately after birth.

Explain the role of colostrum to the calf.

.....
 [2]

(iii) Which hormone is responsible for the enlargement of the udder in Fig. 5.1?

- A follicle stimulating hormone
- B luteinising hormone
- C oestrogen
- D progesterone

Answer **A, B, C** or **D** [1]

(iv) Explain why a cow should be fed a production ration.

.....
 [2]

(b) A dairy cow feeding on dry grass requires 5 dm^3 of water per 50 kg of her live weight. Work out the daily water requirement of a cow weighing 1320 kg. Show your working.

..... dm^3 [2]

(c) Fig. 5.2 shows the digestive system of a calf.

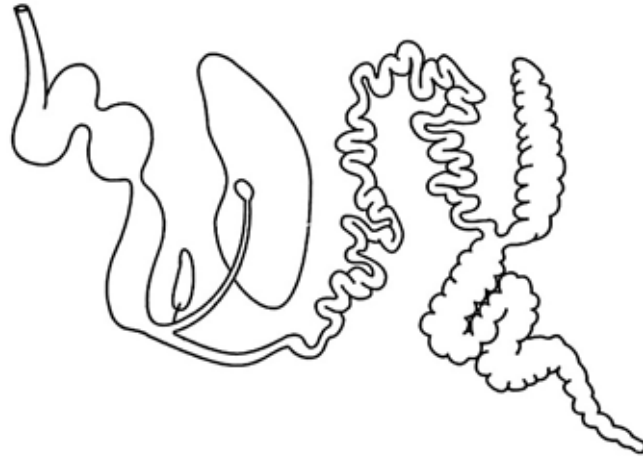


Fig. 5.2

(i) Use label lines to label the following parts on Fig. 5.2.

omasum pancreas small intestine

[3]

(ii) Describe the role of the liver in the digestive system.

.....

.....

.....

.....

[2]

(iii) Suggest why calves up to one week old cannot digest grass or hay.

.....

.....

[1]

[14]

- 6 Fig. 6.1 shows a cow grazing from a contaminated pasture which then died.

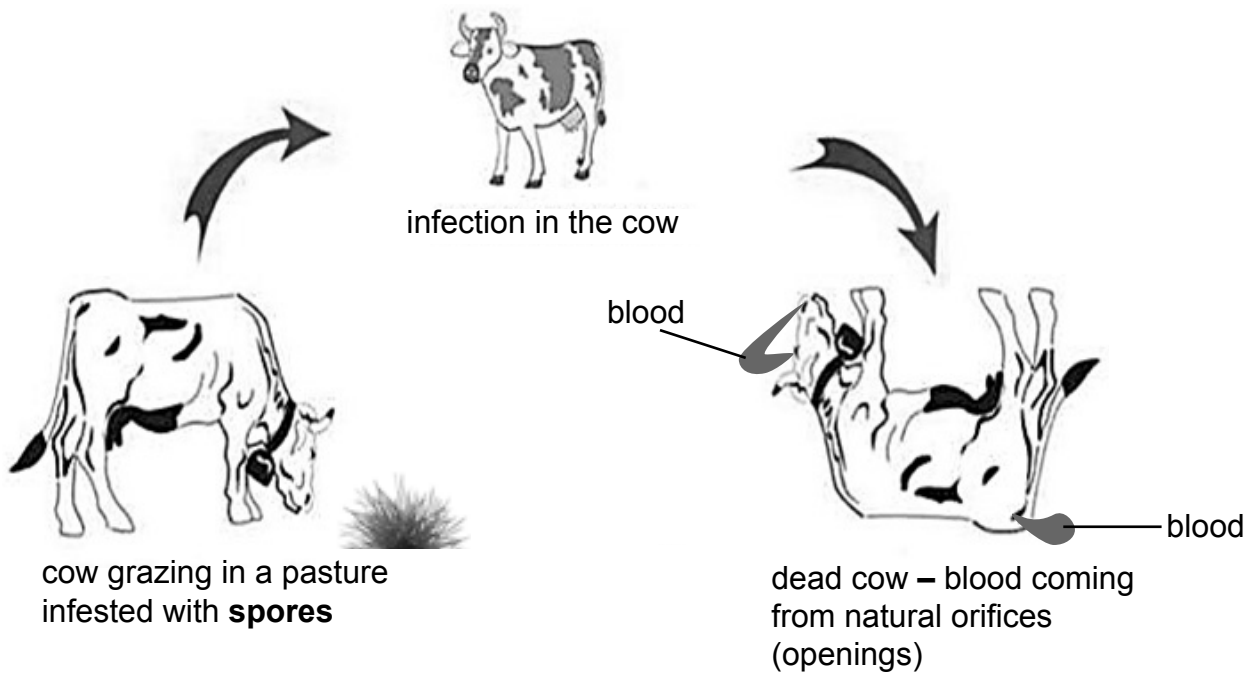


Fig. 6.1

- (a) (i) Name the disease that caused the death of the cow in Fig. 6.1.

..... [1]

- (ii) Describe **two** symptoms of the disease mentioned in (a) (i) other than that shown in Fig. 6.1.

1

2 [2]

- (iii) Suggest how the dead cow in Fig. 6.1 should be safely disposed.

.....

.....

.....

..... [2]

(b) Fig. 6.2 shows a tool used in the prevention of animal diseases.



Fig. 6.2

Which tool is shown in Fig. 6.2?

- A** dosing gun
- B** needle
- C** syringe
- D** vaccinating gun

Answer **A, B, C** or **D** [1]

(c) The following terms are related to the phrases below.

gene **heterozygous** **dominant**
homozygous **chromosome** **genotype**

Choose the correct term and write it below the phrase that describes it.
 Each term may be used once or not at all.

A pair of the same alleles controlling a single characteristic.

.....

Thread-like protein strands found in the nucleus of a cell that
 carries the genes.

.....

Hereditary unit that determines the characteristics of an organism.

.....

[3]

[9]

7 Fig. 7.1 shows a method of treating fencing posts.

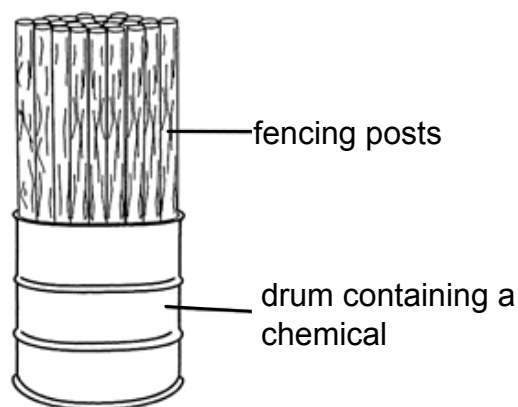


Fig. 7.1

(a) (i) Identify the type of fencing posts in Fig. 7.1.

..... [1]

(ii) Name the chemical used in the drum.

..... [1]

(iii) Explain the need for treating the fencing posts as shown in Fig. 7.1.

.....

.....

.....

.....

..... [3]

(b) Fig. 7.2 shows a water pump.

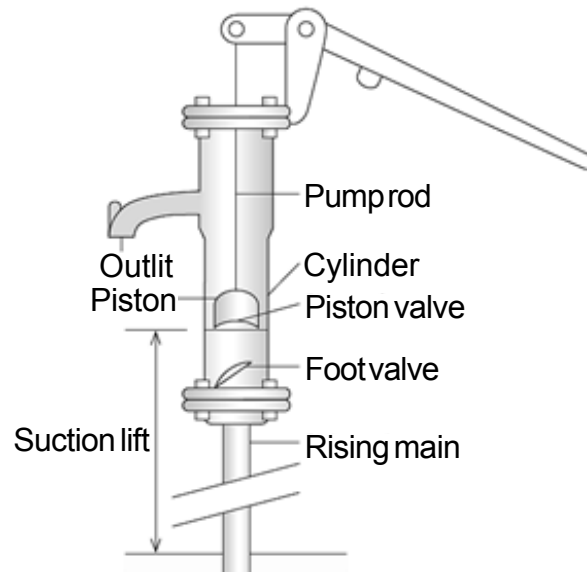


Fig. 7.2

(i) Name the type of water pump shown in Fig. 7.2.

.....

[1]

(ii) Give the functions of a pump rod and foot valve.

pump rod

.....

foot valve

.....

[2]

[8]

- 8 Fig. 8.1 shows an economic principle in trading of wheat flour.

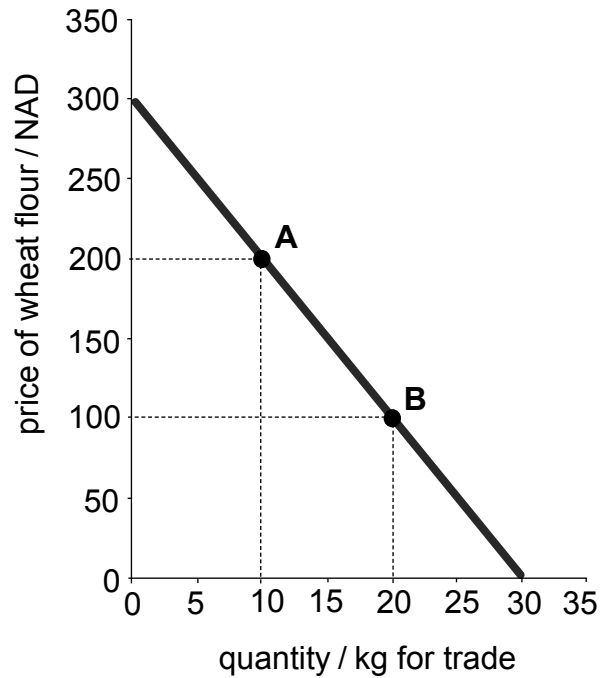


Fig. 8.1

- (a) (i) Name the economic principle illustrated in Fig. 8.1.

[1]

- (ii) Use Fig. 8.1 to determine the quantity of wheat flour when the price is NAD 250.00?

- A** 200 kg
B 100 kg
C 10 kg
D 5 kg

Answer **A, B, C** or **D**

[1]

- (iii) Explain the economic principle in Fig. 8.1 with reference to **A** and **B** on the diagram.

A.....

.....

B.....

.....

[2]

(b) Fig. 8.2 shows farm records **W** and **X**.

W

Assets	NAD	Liabilities	NAD
Cash	5 000	Account payable	8 000
Account receivable	4 500		
Tools and equipment	1 000		

X

CROP:.....	YEAR.....
Spacing:	seed rate:
date planted:	date harvested:
yield:	

Fig. 8.2

Identify the types of farm records labelled **W** and **X** in Fig. 8.2.

W

X

[2]

(c) Discuss the purpose of preparing a budget for a farming enterprise.

.....

.....

.....

.....

.....

[2]

[8]

SECTION B

Answer any **two** questions.

Write your answers on the answer sheets provided at the back of the booklet.

- 9 (a) (i)** Using clear labels and illustrations, describe the process of the water cycle. [5]
- (ii)** Explain how the cutting down of trees may affect the water cycle. [2]
- (b)** Suggest reasons why agriculture is seen as a contributor to the improvement of living conditions of many Namibians in rural communities. [4]
- (c)** Compare conservancies and wildlife councils in the form of a table. [4]
- [15]**
- 10 (a)** Describe shifting cultivation as a method of land use and outline how this method may be helpful in controlling crop pests. [5]
- (b)** Discuss the possible consequences resulting from the loss of biodiversity. [6]
- (c)** Outline **two** advantages and **two** disadvantages of tilling the soil mechanically. [4]
- [15]**
- 11 (a) (i)** Distinguish between soil texture and structure. [2]
- (ii)** Describe one test of soil texture that classifies soil into sand, clay and loam soil. [4]
- (b)** Discuss the effects of over-use of nitrogenous fertilisers on the environment. [5]
- (c)** Suggest ways in which communities can benefit from the tourism industry. [4]
- [15]**
- 12 (a)** Describe the characteristics needed by cattle that are bred for beef. [6]
- (b)** Discuss how the use of lime and fertilisers improves pastures. [5]
- (c)** Suggest the conditions which will result in the Ministry of Environment and Tourism **rejecting** a communal farmers' application to register as a conservancy. [4]
- [15]**

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting or typing. There are no margins, text, or other markings on the page.

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dotted lines, creating a series of uniform gaps for letter height. The entire page is otherwise blank, with no margins, text, or other markings.

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